



Quandela Challenge

Are Quantum PINNs actually doing anything?

ZombieCats

Zombie1 : Chayma

Zombie2 : Iva

Zombie3 : Sahil

Quantum PINNs vs. PINNs

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

PINN

NN + physical intuition

QPINN:

PINN + quantum physics

MAGIC

Quantum physics informed neural networks for multi-variable partial differential equations

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Hugo



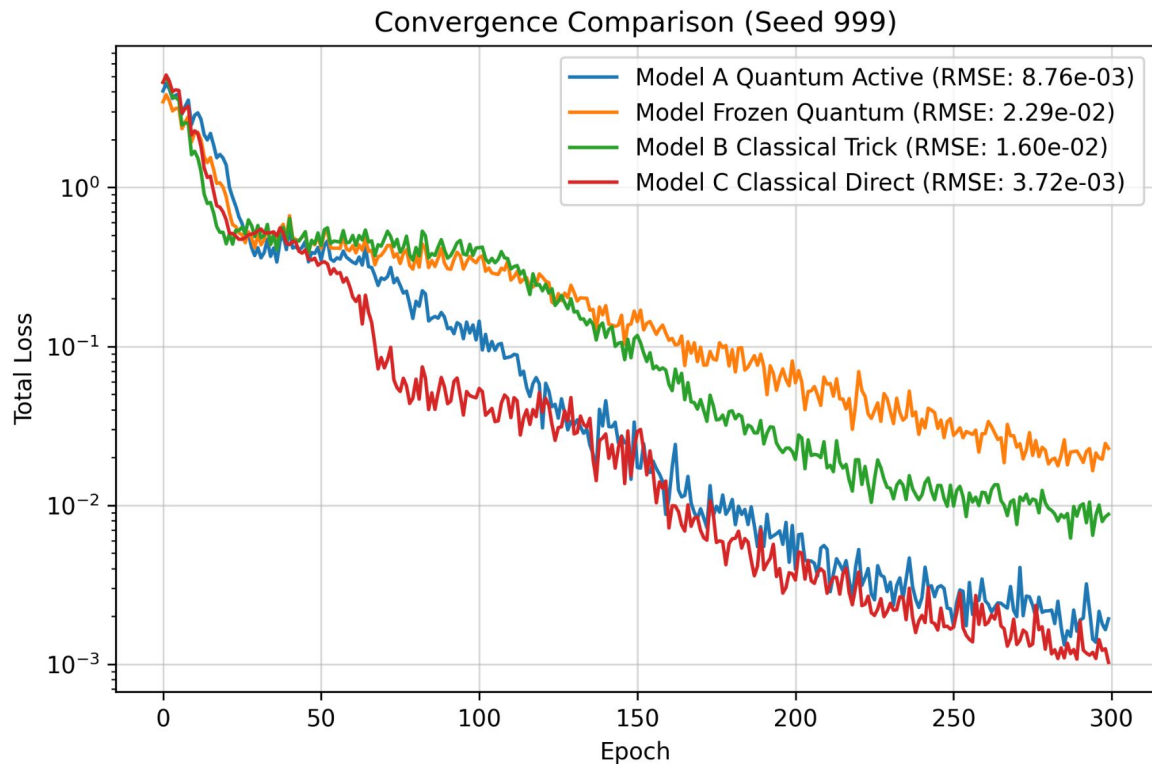
PINNs VS QPINNs: parameter budget

(Q)PINN: ~270 param. :
QPINN follows true
prediction

PINN no convergence!

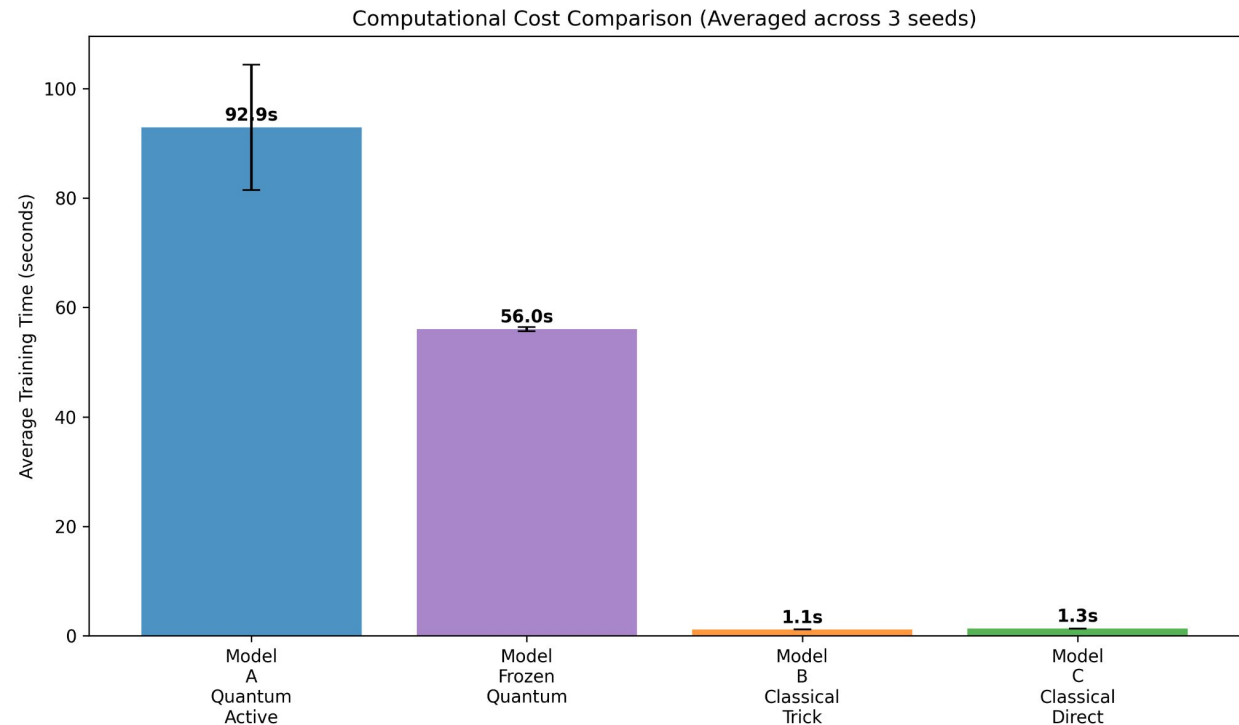
(Q)PINN: ~400 parameters
Success!!!

**This will be our
classical
benchmark!**



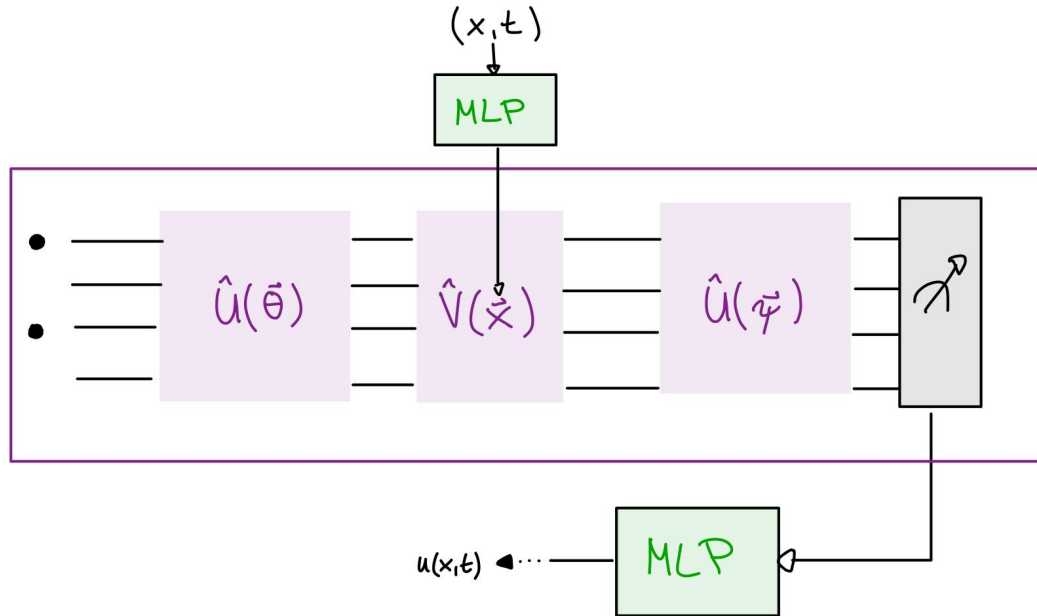
PINNs VS QPINNs: parameter budget with Time

(Q)PINN: ~400 parameters



**Still in the regime of
classical simulability**

Investigating Quantum Resources

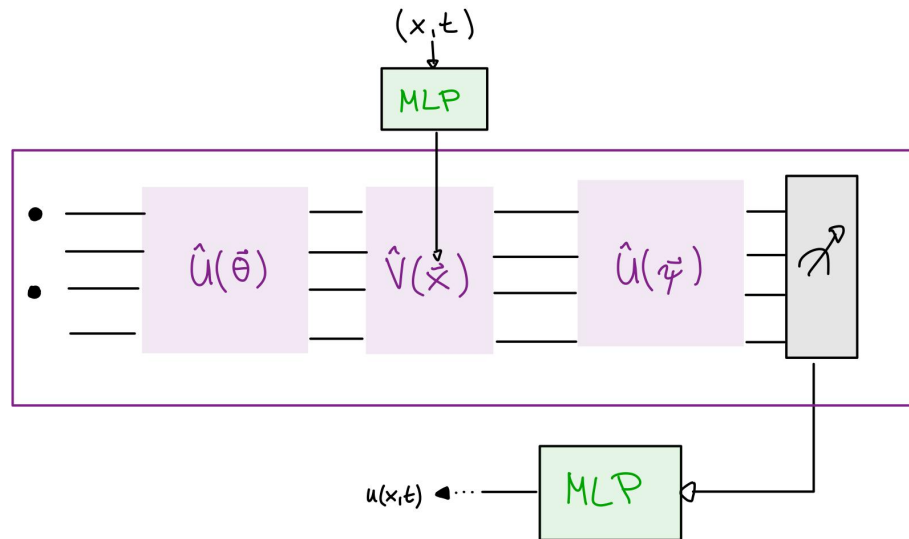


Quantum Resources := # Quantum Parameter

Quantum Resources: circuit 'depth'

QPINN:

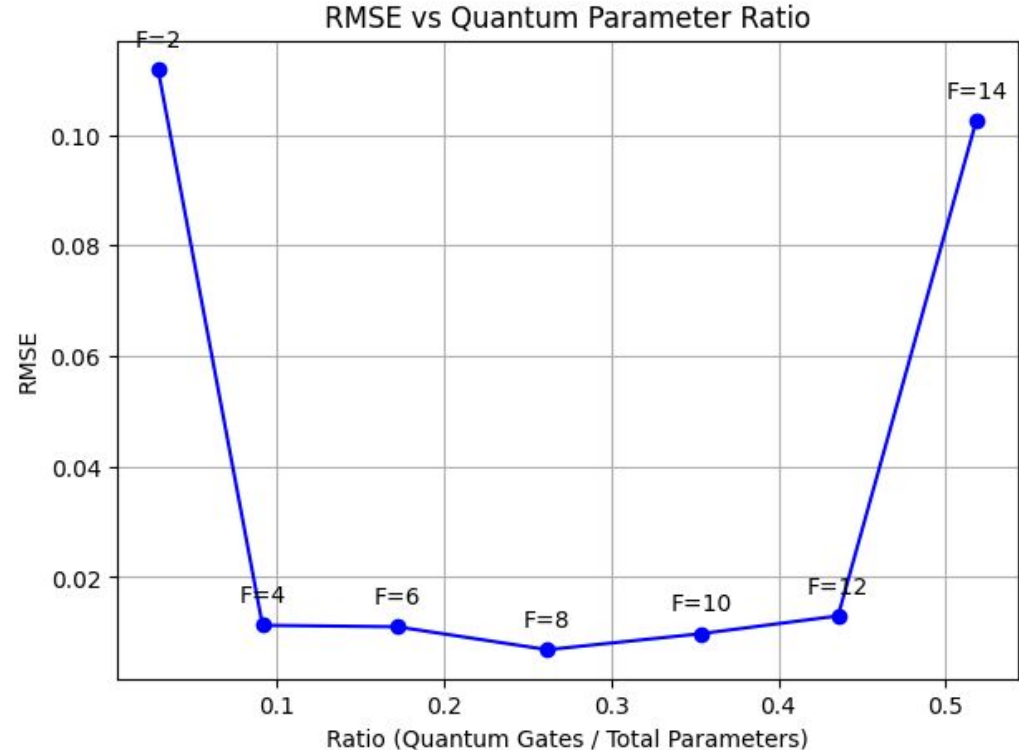
- increase # quantum parameters
- keep # classical parameters constant



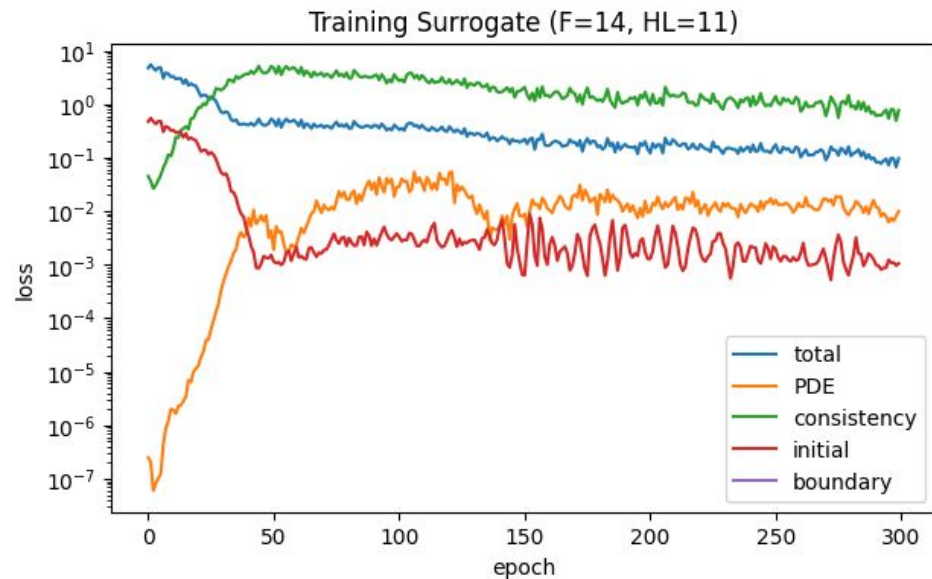
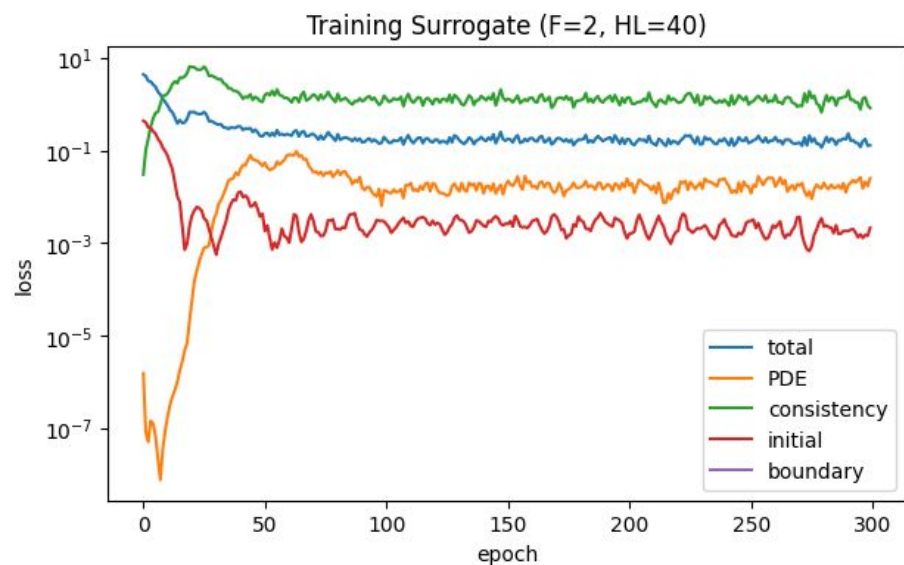
Quantum Resources: circuit 'depth'

Runtime per seed:

25 - 30 min



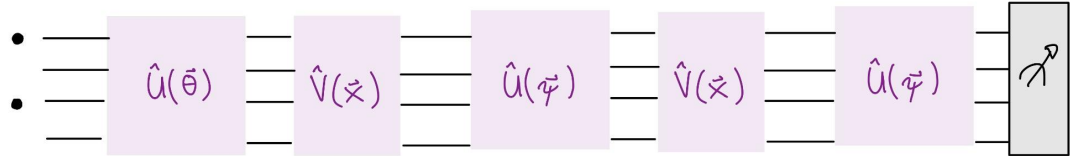
Quantum Resources: circuit 'depth'



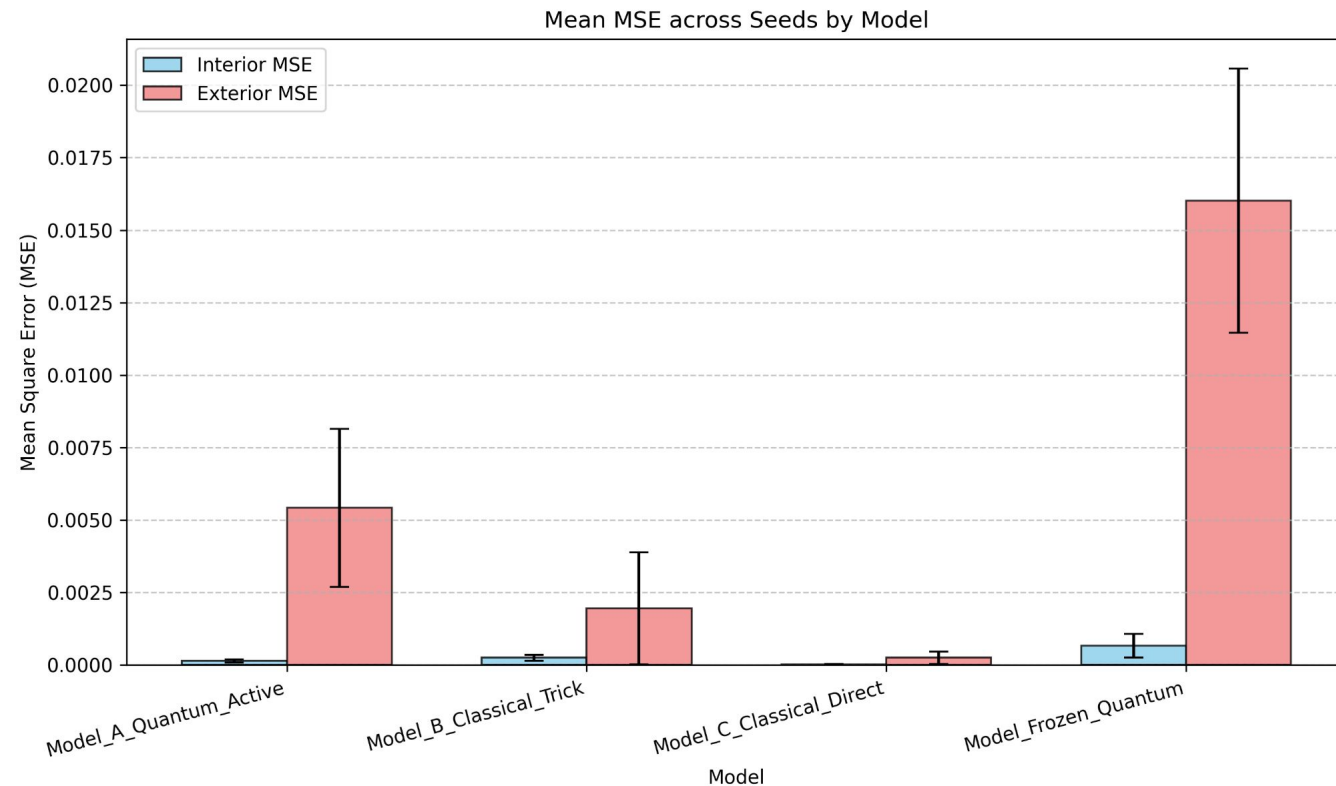
Outlook

Further exploration of 'Quantum':

- implement 'layers', e.g.
- change geometry (U)



PINNs VS QPINNs: parameter budget



(Q)PINN: ~400
parameters, average
across 3 seeds

Summary

Tests

PINN baseline

Quantum resources:

- modify ratio of parameters
- increase both # parameter

Freeze Quantum Layer

Metrics

- MAE
- L_{∞} error
- NMSE
- Loss convergence
- Runtime
- Seed stability